

Unit 8: Special Equations and Classical Models

Prepared by Staples Education

Master Unit Reference Guide: Cheat Sheet, Problems, and Solutions

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Unit 8 at a Glance: Core Sub-Topics

Unit 8 moves from the algebraic theory of first-order ODEs into the classical applied models that drive physics, biology, and economics. It opens with two *special equation types*—the Clairaut equation (singular-solution envelopes) and pursuit/tractrix curves—before building three major applied models: the Navier–Stokes-derived plane Poiseuille velocity profile, the supply-and-demand price-adjustment ODE, and the Lotka–Volterra predator-prey and competition systems. The unit closes with a structured survey of first-order solution methods.

- **8.1 The Clairaut Equation and Singular Solutions** — general solution $y = Cx + f(C)$; singular (envelope) solution from the system $\{y = xp + f(p), 0 = x + f'(p)\}$.
- **8.2 Pursuit Curves and the Tractrix** — tractrix DE $dy/dx = -\sqrt{a^2 - x^2}/x$; general pursuit $xy'' = k\sqrt{1 + (y')^2}$ with speed ratio $k = v_m/v_d$.
- **8.3 Fluid Dynamics and Viscosity** — Newtonian shear $\tau = \mu du/dy$; plane Poiseuille flow $\mu u'' = -G$ with no-slip boundary conditions.
- **8.4 Economic Models, Supply, Demand, and Price** — price ODE $dp/dt = k(D - S)$; equilibrium $p^* = (a + c)/(b + d)$; convergence for $k(b + d) > 0$.
- **8.5 The Lotka–Volterra Predator-Prey Model** — coexistence equilibrium $(c/d, a/b)$; conserved quantity $H = dx - c \ln x + by - a \ln y$.
- **8.6 Competition Models** — two-species carrying-capacity system; nullcline inequalities for stable coexistence versus competitive exclusion.
- **8.7 A Survey of Solution Methods** — decision tree: separable, linear (integrating factor), Bernoulli, exact; classify by structure before solving.

Part I — Condensed Cheat Sheet

8.1 The Clairaut Equation and Singular Solutions

A *Clairaut equation* has the special form $y = xy' + f(y')$. Setting $p = y'$ and differentiating both sides with respect to x yields $(x + f'(p)) dp/dx = 0$, splitting into two branches whose solutions are the general family of straight lines and the singular envelope.

Clairaut Equation: General and Singular Solutions

For $y = xp + f(p)$ with $p = y'$:

General solution: $y = Cx + f(C)$ ($p = C = \text{const}$),

Singular solution (envelope):
$$\begin{cases} y = xp + f(p) \\ 0 = x + f'(p) \end{cases}$$

Eliminate p from this system to obtain the envelope: a curve tangent to every line in the general family. The singular solution satisfies the ODE but *cannot* be obtained from the general solution by any choice of C .

Envelope Elimination Procedure

1. Write the equation as $F(x, y, C) = y - Cx - f(C) = 0$.
2. Differentiate with respect to C : $\partial F / \partial C = -x - f'(C) = 0$, giving $x = -f'(C)$.
3. Solve for $C = C(x)$ and substitute back into $F = 0$ to obtain the envelope curve $y = y(x)$.
4. Verify: the envelope must satisfy the original ODE directly.

Common Pitfall: Singular Solution Is Not a Particular Solution

The singular solution is *not* a member of the general family $y = Cx + f(C)$ for any finite C . Derive it exclusively via the envelope elimination system; attempting to recover it by substituting a value of C will always fail.

8.2 Pursuit Curves and the Tractrix

A *pursuit curve* traces the path of a pursuer who always aims directly at a moving target. The tractrix is the classical limiting case in which the pursuer is dragged at constant distance a from a target walking along a line.

Tractrix Differential Equation and Solution

Target on the y -axis, pursuer starts at $(a, 0)$, equal speeds, string length a :

$$\frac{dy}{dx} = -\frac{\sqrt{a^2 - x^2}}{x}, \quad 0 < x \leq a, \quad y(a) = 0.$$

Solution (via $x = a \sin \theta$):

$$y(x) = a \ln \left(\frac{a + \sqrt{a^2 - x^2}}{x} \right) - \sqrt{a^2 - x^2}.$$

General Pursuit Equation

Dog at origin pursues owner on line $x = c$ at speed v_m ; dog speed $v_d > v_m$; $k = v_m/v_d < 1$:

$$x y'' = k \sqrt{1 + (y')^2}, \quad y(c) = 0, \quad y'(c) = 0.$$

Solution ($k \neq 1$):

$$y(x) = \frac{1}{2} \left[\frac{x^{k+1}}{(k+1)c^k} - \frac{c^k x^{1-k}}{1-k} \right] + \frac{ck}{1-k^2}.$$

Capture height: $y(0) = ck/(1-k^2)$.

Common Pitfall: Direction of the Derivative Sign

In the tractrix, dy/dx is *negative* because x decreases as the pursuer is dragged toward the y -axis. Always verify the geometric sign from the tangent-segment picture before writing the ODE.

8.3 Fluid Dynamics and Viscosity

Newtonian viscosity: $\tau = \mu du/dy$. Plane Poiseuille flow reduces the momentum balance to a second-order constant-coefficient ODE with a parabolic solution.

Plane Poiseuille Velocity Profile

$$\mu u''(y) = -G, \quad u(0) = u(h) = 0:$$

$$u(y) = \frac{G}{2\mu} y(h-y), \quad y_{\max} = \frac{h}{2}, \quad u_{\max} = \frac{Gh^2}{8\mu}.$$

Shear stress: $\tau(y) = \frac{G}{2}(h-2y)$, linear, zero at the centerline.

2D Incompressible Continuity and Stream Function

$\partial u/\partial x + \partial v/\partial y = 0$. Given u : integrate $\partial v/\partial y = -\partial u/\partial x$ in y . Stream function: $u = \partial \psi/\partial y$, $v = -\partial \psi/\partial x$; streamlines are $\psi(x, y) = C$.

Common Pitfall: Both No-Slip Conditions Are Required

The Poiseuille ODE produces a quadratic with two free constants. Both $u(0) = 0$ and $u(h) = 0$ must be applied; using only one gives a wrong family.

8.4 Economic Models, Supply, Demand, and Price

Price adjusts at a rate proportional to excess demand. With linear demand and supply the ODE is first-order linear, and the equilibrium price is globally stable.

Price Adjustment ODE and Equilibrium

$D = a - bp$, $S = -c + dp$ ($a, b, c, d > 0$):

$$\frac{dp}{dt} = k[(a + c) - (b + d)p], \quad p(t) = p^* + (p_0 - p^*)e^{-k(b+d)t}, \quad p^* = \frac{a + c}{b + d}.$$

Convergence: $k(b + d) > 0$ always.

Solving the Price ODE via Integrating Factor

1. Write $dp/dt + k(b + d)p = k(a + c)$.
2. $\mu(t) = e^{k(b+d)t}$; multiply and integrate.
3. Apply $p(0) = p_0$: $C = p_0 - p^*$.

8.5 The Lotka–Volterra Predator-Prey Model

The Lotka–Volterra system produces closed phase-plane orbits guaranteed by the first integral $H(x, y)$.

Lotka–Volterra System and Conserved Quantity

$\dot{x} = ax - bxy$, $\dot{y} = -cy + dxy$ ($a, b, c, d > 0$):

$$E_0 = (0, 0), \quad E^* = (c/d, a/b).$$

$$H(x, y) = dx - c \ln x + by - a \ln y = \text{const} \quad (\text{minimum at } E^*).$$

Deriving the Conserved Quantity

1. Form $dy/dx = \dot{y}/\dot{x} = y(-c + dx)/[x(a - by)]$.
2. Separate: $(a/y - b) dy = (-c/x + d) dx$.
3. Integrate: $a \ln y - by = -c \ln x + dx + \text{const}$.
4. Rearrange: $H = dx - c \ln x + by - a \ln y$.
5. Verify $\nabla H = 0$ at E^* and Hessian positive definite.

Common Pitfall: The Coexistence Equilibrium Is a Center

$(c/d, a/b)$ is a *center* (closed orbits), not a stable spiral. Perturbations do not damp out in the pure Lotka–Volterra model; adding saturation converts the center to a stable spiral.

8.6 Competition Models

Two-species competition: the outcome depends on the relative positions of the x - and y -nullclines.

Competition System and Interior Equilibrium

$$\dot{x} = r_1x(1 - (x + \alpha y)/K_1), \quad \dot{y} = r_2y(1 - (y + \beta x)/K_2).$$

$$E_3 = (x^*, y^*), \quad x^* = \frac{K_1 - \alpha K_2}{1 - \alpha\beta}, \quad y^* = \frac{K_2 - \beta K_1}{1 - \alpha\beta}.$$

Nullclines: $x + \alpha y = K_1$ and $\beta x + y = K_2$.

Coexistence vs. Exclusion: Nullcline Intercept Test

- **Stable coexistence** ($\alpha\beta < 1$): $K_1 < K_2/\beta$ and $K_2 < K_1/\alpha$.
- **Bistable exclusion** ($\alpha\beta > 1$): $K_1 > K_2/\beta$ and $K_2 > K_1/\alpha$.
- **x wins**: $K_1 > K_2/\beta$ and $K_2 < K_1/\alpha$.
- **y wins**: $K_1 < K_2/\beta$ and $K_2 > K_1/\alpha$.

Common Pitfall: Interior Equilibrium Feasibility

E_3 is biologically meaningful only when both $x^* > 0$ and $y^* > 0$. Check the signs of numerators and denominator before drawing conclusions.

Unit 8 Key Takeaway — Part I

Clairaut equations split into a **general line family** $y = Cx + f(C)$ and a **singular envelope** from $x + f'(p) = 0$. The Poiseuille profile $u = (G/2\mu)y(h - y)$ and price path $p(t) = p^* + (p_0 - p^*)e^{-k(b+d)t}$ follow from linear BVPs. Lotka–Volterra orbits are level sets of $H = dx - c \ln x + by - a \ln y$; competition outcomes follow from $\alpha\beta < 1$ (coexistence) versus $\alpha\beta > 1$ (exclusion).

Part II — Review Guide: Practice Problems

Work the following ten problems in order. Problems 1–2 develop the Clairaut equation: general straight-line family and singular envelope elimination (Module 8.1). Problems 3–4 derive and solve the tractrix ODE and the general pursuit equation with a speed-ratio parameter (Module 8.2). Problem 5 solves the plane Poiseuille velocity profile with no-slip boundary conditions, and Problem 6 applies 2D incompressible continuity to find the stream function (Module 8.3). Problem 7 solves the linear price-adjustment ODE and identifies the equilibrium (Module 8.4). Problem 8 analyzes the full Lotka–Volterra predator-prey system including the conserved quantity H (Module 8.5). Problem 9 classifies coexistence versus competitive exclusion in a two-species competition model via nullcline inequalities (Module 8.6). Problem 10 closes with a four-part classification and solution survey (Module 8.7).

1. Find the general solution and the singular solution of the Clairaut equation $y = xy' + (y')^2$. Verify that the singular solution is the envelope of the general straight-line family.
2. For the Clairaut equation $y = xy' - \ln(y')$ (with $y' > 0$), find the one-parameter family of straight-line solutions and the singular solution. Confirm that the singular solution satisfies the ODE directly.
3. A target moves from the origin up the positive y -axis. A pursuer starts at $(a, 0)$ and is always connected to the target by a taut string of length a , with both moving at the same speed. Derive the tractrix differential equation

$$\frac{dy}{dx} = -\frac{\sqrt{a^2 - x^2}}{x}, \quad y(a) = 0,$$

from the constant-length tangent-line geometry, then solve it for $y(x)$ using the substitution $x = a \sin \theta$.

4. A dog at the origin always runs directly toward its owner, who walks along $x = c$ at speed v_m . The dog runs at speed $v_d > v_m$ with $k = v_m/v_d$. Derive $xy'' = k\sqrt{1 + (y')^2}$, $y(c) = 0$, $y'(c) = 0$. Set $p = y'$, solve the separable equation in p , integrate to find $y(x)$, and state the finite capture height $y(0)$.
5. Solve $\mu u''(y) = -G$, $u(0) = u(h) = 0$ for the velocity profile $u(y)$. State the location and value of the maximum velocity.
6. Given $u = x^2 - y^2$ for a 2D incompressible steady flow, find the most general $v(x, y)$ consistent with $\partial u/\partial x + \partial v/\partial y = 0$. For $g(x) = 0$, determine the stream function $\psi(x, y)$ and write the streamline equation.
7. Price obeys $dp/dt = k(D - S)$ with $D = a - bp$, $S = -c + dp$ ($a, b, c, d, k > 0$). Write the ODE, solve via integrating factor with $p(0) = p_0$, find p^* , and state the convergence condition.
8. For $\dot{x} = ax - bxy$, $\dot{y} = -cy + dxy$ ($a, b, c, d > 0$): interpret each parameter; find both equilibria; derive the conserved quantity $H(x, y) = dx - c \ln x + by - a \ln y$; verify H is minimized at $(c/d, a/b)$.

9. For the competition system with parameters $r_1, r_2, K_1, K_2, \alpha, \beta$: find all four equilibria; write the nullclines; state the inequalities governing stable coexistence, bistable exclusion, and single-species dominance; derive (x^*, y^*) explicitly.
10. Classify and solve completely: (a) $dy/dx = x(1 + y^2)/(1 + x^2)$; (b) $dy/dx + (2/x)y = x^3$; (c) $dy/dx + y = xy^3$; (d) $(2xy + \cos x)dx + (x^2 - \sin y)dy = 0$.
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Part III — Complete Solutions

Solution 1. Set $p = y'$. The Clairaut equation is $y = xp + p^2$.

Differentiate both sides with respect to x :

$$p = p + x \frac{dp}{dx} + 2p \frac{dp}{dx} = p + (x + 2p) \frac{dp}{dx}.$$

Cancel p :

$$(x + 2p) \frac{dp}{dx} = 0.$$

Case A: $dp/dx = 0 \Rightarrow p = C$. **General solution:** $y = Cx + C^2$.

Case B: $x + 2p = 0 \Rightarrow p = -x/2$. Substitute:

$$y = x \left(-\frac{x}{2} \right) + \left(-\frac{x}{2} \right)^2 = -\frac{x^2}{2} + \frac{x^2}{4} = -\frac{x^2}{4}.$$

Envelope check: $F = y - Cx - C^2$; $\partial F / \partial C = -x - 2C = 0$, so $C = -x/2$. Substituting:

$$y + \frac{x^2}{2} - \frac{x^2}{4} = y + \frac{x^2}{4} = 0 \implies y = -\frac{x^2}{4}.$$

Direct check: $y' = -x/2$, $(y')^2 = x^2/4$, $xy' + (y')^2 = -x^2/2 + x^2/4 = -x^2/4 = y$. ✓

Solution 2. Set $p = y' > 0$. The equation is $y = xp - \ln p$.

Differentiate:

$$p = p + \left(x - \frac{1}{p} \right) \frac{dp}{dx} \implies \left(x - \frac{1}{p} \right) \frac{dp}{dx} = 0.$$

Case A: $p = C$. **General solution:** $y = Cx - \ln C$.

Case B: $p = 1/x$. Substitute:

$$y = x \cdot \frac{1}{x} - \ln \frac{1}{x} = 1 + \ln x.$$

Singular solution: $y = 1 + \ln x$.

Envelope: $F = y - Cx + \ln C$; $\partial F / \partial C = -x + 1/C = 0$, so $C = 1/x$. Then $y - 1 - \ln x = 0$. ✓

Direct check: $y' = 1/x$; $xy' - \ln y' = 1 + \ln x = y$. ✓

Solution 3. At (x, y) the string has horizontal leg x and hypotenuse a , so vertical leg $= \sqrt{a^2 - x^2}$. Since x decreases as the pursuer is dragged:

$$\frac{dy}{dx} = -\frac{\sqrt{a^2 - x^2}}{x}.$$

Substitute $x = a \sin \theta$, $dx = a \cos \theta d\theta$, $\sqrt{a^2 - x^2} = a \cos \theta$:

$$\int \frac{\sqrt{a^2 - x^2}}{x} dx = a \int \frac{\cos^2 \theta}{\sin \theta} d\theta = a \int (\csc \theta - \sin \theta) d\theta = a [\ln |\csc \theta - \cot \theta| + \cos \theta].$$

Back-substitute ($\csc \theta = a/x$, $\cot \theta = \sqrt{a^2 - x^2}/x$, $\cos \theta = \sqrt{a^2 - x^2}/a$):

$$= a \ln \left(\frac{a - \sqrt{a^2 - x^2}}{x} \right) + \sqrt{a^2 - x^2}.$$

So $y = -a \ln((a - \sqrt{a^2 - x^2})/x) - \sqrt{a^2 - x^2} + K$. Applying $y(a) = 0$ gives $K = 0$. Rationalizing:

$$y(x) = a \ln\left(\frac{a + \sqrt{a^2 - x^2}}{x}\right) - \sqrt{a^2 - x^2}.$$

Solution 4. Tangency condition at dog position (x, y) with owner at (c, Y) : $Y - y = (c - x)y'$. Differentiate in x : $dY/dx = (c - x)y''$. Speed ratio: $dY/dx = k\sqrt{1 + (y')^2}$. Equating: $(c - x)y'' = k\sqrt{1 + (y')^2}$, i.e. $xy'' = k\sqrt{1 + (y')^2}$ (after shifting).

Let $p = y'$; separate $dp/\sqrt{1 + p^2} = k dx/x$:

$$\operatorname{arcsinh}(p) = k \ln x + C_1 \implies p + \sqrt{1 + p^2} = A x^k.$$

$y'(c) = 0$: $A = c^{-k}$. Using $(\sqrt{1 + p^2} + p)(\sqrt{1 + p^2} - p) = 1$:

$$2p = (x/c)^k - (x/c)^{-k} \implies y' = \frac{1}{2} \left[(x/c)^k - (x/c)^{-k} \right].$$

Integrate ($k \neq 1$):

$$y = \frac{1}{2} \left[\frac{x^{k+1}}{(k+1)c^k} - \frac{c^k x^{1-k}}{1-k} \right] + C_2.$$

$y(c) = 0$: $C_2 = ck/(1 - k^2)$.

Solution: $y(x) = \frac{1}{2} [x^{k+1}/((k+1)c^k) - c^k x^{1-k}/(1-k)] + ck/(1 - k^2)$. Capture height: $y(0) = ck/(1 - k^2)$.

Solution 5. Integrate $\mu u'' = -G$ twice:

$$\mu u'(y) = -Gy + C_1, \quad \mu u(y) = -\frac{G}{2}y^2 + C_1y + C_2.$$

$u(0) = 0 \Rightarrow C_2 = 0$. $u(h) = 0 \Rightarrow C_1 = Gh/2$.

$$u(y) = \frac{G}{2\mu} y(h - y).$$

$u'(y) = (G/2\mu)(h - 2y) = 0$ at $y = h/2$:

$$u_{\max} = \frac{G}{2\mu} \cdot \frac{h}{2} \cdot \frac{h}{2} = \frac{Gh^2}{8\mu}.$$

Solution 6. $\partial u/\partial x = 2x$, so $\partial v/\partial y = -2x$. Integrate in y :

$$v(x, y) = -2xy + g(x).$$

With $g = 0$, find ψ from $\partial\psi/\partial y = x^2 - y^2$:

$$\psi = x^2y - \frac{y^3}{3} + h(x).$$

Check $-\partial\psi/\partial x = -2xy - h'(x) = v = -2xy$, so $h' = 0$, $h = \text{const}$. Streamlines:

$$x^2y - \frac{y^3}{3} = k.$$

Solution 7.

$$\frac{dp}{dt} = k[(a+c) - (b+d)p] \implies \frac{dp}{dt} + k(b+d)p = k(a+c).$$

$\mu = e^{k(b+d)t}$; integrate:

$$p(t) = \frac{a+c}{b+d} + \left(p_0 - \frac{a+c}{b+d}\right) e^{-k(b+d)t}.$$

$p^* = (a+c)/(b+d)$. Convergence: $k(b+d) > 0$ (automatic). ✓

Solution 8. *Parameters:* a = prey growth rate; b = predation coefficient; c = predator death rate; d = conversion efficiency.

Equilibria: $\dot{x} = 0 \implies x = 0$ or $y = a/b$; $\dot{y} = 0 \implies y = 0$ or $x = c/d$:

$$E_0 = (0, 0), \quad E^* = (c/d, a/b).$$

Conserved quantity. $dy/dx = y(-c+dx)/[x(a-by)]$; separate:

$$\left(\frac{a}{y} - b\right) dy = \left(-\frac{c}{x} + d\right) dx.$$

Integrate:

$$a \ln y - by = -c \ln x + dx + \text{const} \implies H = dx - c \ln x + by - a \ln y = \text{const}.$$

$\partial H/\partial x = d - c/x = 0 \implies x = c/d$; $\partial H/\partial y = b - a/y = 0 \implies y = a/b$. Hessian: $\partial^2 H/\partial x^2 = c/x^2 > 0$, $\partial^2 H/\partial y^2 = a/y^2 > 0$, cross term = 0: positive definite. E^* is the unique minimum of H .

Solution 9. *Equilibria:* $(0, 0)$, $(K_1, 0)$, $(0, K_2)$, and

$$(x^*, y^*) = \left(\frac{K_1 - \alpha K_2}{1 - \alpha\beta}, \frac{K_2 - \beta K_1}{1 - \alpha\beta}\right).$$

Derivation: $x + \alpha y = K_1$ and $\beta x + y = K_2$; substitute $x = K_1 - \alpha y$ into the second:

$$\beta K_1 + y(1 - \alpha\beta) = K_2 \implies y^* = \frac{K_2 - \beta K_1}{1 - \alpha\beta}, \quad x^* = \frac{K_1 - \alpha K_2}{1 - \alpha\beta}.$$

Nullclines: $x + \alpha y = K_1$ (axes intercepts $(K_1, 0)$ and $(0, K_1/\alpha)$); $\beta x + y = K_2$ (axes intercepts $(K_2/\beta, 0)$ and $(0, K_2)$).

Outcome inequalities:

Coexistence : $K_1 < K_2/\beta$ and $K_2 < K_1/\alpha$ ($\alpha\beta < 1$).

Bistable exclusion : $K_1 > K_2/\beta$ and $K_2 > K_1/\alpha$ ($\alpha\beta > 1$).

x wins : $K_1 > K_2/\beta$ and $K_2 < K_1/\alpha$. y wins : $K_1 < K_2/\beta$ and $K_2 > K_1/\alpha$.

Solution 10.

(a) *Separable.*

$$\frac{dy}{1+y^2} = \frac{x dx}{1+x^2} \implies \arctan y = \frac{1}{2} \ln(1+x^2) + C \implies y = \tan\left(\frac{1}{2} \ln(1+x^2) + C\right).$$

(b) *Linear, integrating factor.* $\mu = x^2$; $d(x^2 y)/dx = x^5$; integrate:

$$x^2 y = \frac{x^6}{6} + C \implies y = \frac{x^4}{6} + \frac{C}{x^2}.$$

(c) Bernoulli, $n = 3$. Let $v = y^{-2}$: $v' - 2v = -2x$. $\mu = e^{-2x}$:

$$\frac{d}{dx}(v e^{-2x}) = -2x e^{-2x}.$$

Integrate by parts: $\int -2x e^{-2x} dx = x e^{-2x} + e^{-2x}/2$:

$$v = x + \frac{1}{2} + C e^{2x} \implies y^2 = \frac{1}{x + \frac{1}{2} + C e^{2x}}.$$

(d) Exact. $M_y = 2x = N_x$. Integrate M in x :

$$F = x^2 y + \sin x + g(y); \quad F_y = x^2 + g'(y) = x^2 - \sin y \implies g(y) = \cos y.$$

$$x^2 y + \sin x + \cos y = C.$$

Unit 8 Key Takeaway

Every Clairaut equation $y = xy' + f(y')$ yields a **general line family** $y = Cx + f(C)$ and a **singular envelope** obtained by eliminating p from $x + f'(p) = 0$. Pursuit geometry converts tangency and speed constraints into separable second-order ODEs. The Poiseuille profile $u = (G/2\mu) y(h - y)$ and the price path $p(t) = p^* + (p_0 - p^*)e^{-k(b+d)t}$ both follow from linear BVPs. In Lotka–Volterra dynamics the **conserved quantity** $H = dx - c \ln x + by - a \ln y$ organizes all phase-plane orbits as closed curves around the coexistence equilibrium $(c/d, a/b)$; in two-species competition the entire outcome is decided by $\alpha\beta < 1$ (**stable coexistence**) versus $\alpha\beta > 1$ (**competitive exclusion**).